

Chapter 2: Account and Security Administration, and Access Control

In Linux, a single user account generally belongs to a single user. The permissions for this user account determine the **level of access** the user will have while using the system.

Types of user accounts

1. **Super User:** This account is also known as the **root account**. It has all the permissions and can run all the commands without any restriction. The keyword **sudo** is often used to run a command as the root user.
2. **Normal User:** The **general users** of the system are assigned this account. This account has restricted access to the commands and files of the system. The root user can create such an account, modify its permissions or delete it.
3. **System User:** This is the type of account that is created only for a **specific purpose** or software. For example, a mail account.

2.1 Account and security administration

Account and security administration involves managing user accounts, groups, and implementing security measures to protect the system.

2.1.1 User and group concepts, and user private group scheme

User and Group concept

In a computer system, users are individuals who **interact with the system**, while groups are **collections of users** with similar access requirements. Here are some important concepts related to users and groups:

- ✓ **User accounts:** User accounts are created for individual users to access the system. Each user is assigned a unique **username** and **password**, which they use to authenticate themselves.
- ✓ **Group accounts:** Group accounts are created to manage access permissions collectively for **a group of users**. Users who are part of a group inherit the access rights and permissions associated with that group.

User identification (UID) and group identification (GID): Each user and group is assigned a unique identification number (UID and GID, respectively) by **the system**. These IDs are used internally by the system to identify and manage users and groups.

User Private Group Scheme:

The **User Private Group** (UPG) scheme is a **common approach** in Unix-like operating systems (such as Linux) to manage user and group administration. In this scheme:

- ✓ Each user is associated with a primary group, which has the same name as the username and the same GID.
- ✓ A private group is created for each user, with the same name as the primary group and a unique GID.
- ✓ The private group serves as a secondary group for the user and allows for better file and directory permissions management.

The UPG scheme helps enhance security by isolating each user's files and directories within their private group, preventing unauthorized access by other users.

2.1.2 User administration, modifying accounts and group administration

User administration involves **creating**, **modifying**, and **managing** user accounts. Some common tasks in user administration include:

- ✓ **Creating user accounts:** Administrators can create new user accounts by specifying a username, password, and other relevant details.
- ✓ **Modifying user accounts:** Administrators can modify user account settings such as the account name, password, home directory, shell, and resource limits.
- ✓ **Disabling or deleting user accounts:** In certain cases, user accounts may need to be disabled or deleted to revoke access.

Administrators typically use command-line tools or graphical interfaces provided by the operating system to perform user administration tasks.

Group Administration

Group administration involves creating and managing groups to **simplify access control**. Key tasks in group administration include:

- ✓ Creating groups: Administrators can create new groups and assign them a unique GID.
- ✓ Modifying groups: Admins can modify group settings such as the group name or GID.
- ✓ Adding or removing users from groups: Users can be added or removed from groups to manage their access permissions.

Group administration allows for the efficient management of access rights by assigning permissions to groups rather than individual users, simplifying the overall security management process.

2.1.3 Password aging and default user files

To **enhance security**, systems often implement **password aging policies** and provide default user files. Here's what these concepts involve:

- ✓ **Password aging**: Password aging refers to the practice of enforcing periodic password changes. It involves setting a maximum password age, after which users are prompted to change their passwords.
- ✓ **Default user files**: Default user files are preconfigured files or directories provided to users when their accounts are created. These files often include initial configuration settings, default preferences, and skeleton files for user-specific customization.

Password aging helps to ease the risk of compromised passwords by ensuring that **users regularly** update their passwords. **Default user files** provide a consistent starting point for user accounts, ensuring that users have necessary files and configurations readily available when they log in for the first time.

Overall, **effective account** and **security administration**, including user and group management, password aging, and default user files, contribute to maintaining the integrity, confidentiality, and availability of computer systems and protecting against unauthorized access.

2.2 Managing files and folder permission

Managing files and folder permissions, **controlling access to files**, and **managing disk quotas** are **important** aspects of system administration that help ensure proper data security, access control, and resource management.

2.2.1 Managing file ownership

File and folder permissions determine who can access, modify, and execute files and folders on a system. These permissions are typically represented by a combination of read (r), write (w), and execute (x) permissions, which can be assigned to three different groups: the owner of the file, the group associated with the file, and others (everyone else).

To manage file and folder permissions, **administrators** can use commands like `chmod` (change mode) in Unix-like systems or the Security tab in Windows systems. Key tasks include:

- ✓ **Setting permissions:** Administrators can assign appropriate permissions to files and folders, specifying which users or groups can read, write, or execute them.
- ✓ **Modifying permissions:** Administrators can modify existing permissions to grant or revoke access rights as needed.
- ✓ **Viewing permissions:** Administrators can view the current permissions assigned to files and folders to ensure proper access control.

2.2.2 Controlling access to files

File ownership refers to determining the user and group associated with a file or folder. Administrators can manage file ownership to control access and permissions. Key tasks include:

- ✓ **Changing file ownership:** Administrators can change the owner and group associated with a file using commands like `chown` (change owner) or `chgrp` (change group) in Unix-like systems.
- ✓ **Transferring ownership:** Administrators can transfer file ownership from one user to another, ensuring that the appropriate user has control over the file and its permissions.

Controlling access to files involves setting up **access control lists** (ACLs) or managing file permissions to restrict or grant access to specific users or groups. Key tasks include:

- ✓ **Setting up ACLs:** ACLs allow administrators to define fine-grained access control, specifying access permissions for individual users or groups on a file or folder basis.
- ✓ **Assigning special permissions:** Administrators can assign special permissions such as `setuid`, `setgid`, or sticky bits to files or folders to control access and execution behavior.

Controlling access to files ensures that only **authorized** users or groups can access **sensitive data**, protecting against unauthorized **disclosure** or **modification**.

2.2.3 Managing disk quotas

Disk quotas enable administrators to **limit the amount of disk space** individual users or groups can consume on a file system. This helps in managing disk usage and preventing any individual or group from monopolizing system resources. Key tasks include:

- ✓ **Enabling disk quotas:** Administrators can enable disk quotas on specific file systems to enforce usage limits.
- ✓ **Setting quota limits:** Administrators can set quota limits for individual users or groups, specifying the maximum amount of disk space they can consume.

- ✓ **Monitoring disk usage:** Administrators can monitor disk usage by users or groups to ensure compliance with the specified quotas.

By implementing disk quotas, administrators can **prevent excessive disk usage**, ensure **fair resource allocation**, and avoid potential system performance issues.

Proper management of **files and folder permissions**, **file ownership**, **access control**, and **disk quotas** is crucial for maintaining data security, controlling access, and effectively managing system resources.